

Problem set 1 solution

Anakawee Chujinda

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1 The real numbers

1.1 Ordered fields and completeness

A quick reminder of notation: $\mathbb{N}$, $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$, $\mathbb{C}$ are the usual number systems, and $\varepsilon > 0$ always denotes a real number.

Definition 1.1. Let $S \subseteq \mathbb{R}$ be nonempty. We say $b \in \mathbb{R}$ is an **upper bound** of S if $x \leq b$ for every $x \in S$. The **supremum** $\sup S$ is the least upper bound of S , when one exists.

Example 1.2 (Rationals are not complete)

The set $S = \{q \in \mathbb{Q} \mid q^2 < 2\}$ is bounded above in $\mathbb{Q}$ but has no least upper bound in $\mathbb{Q}$.

Theorem 1.3 (Archimedean property)

If $x, y \in \mathbb{R}$ and $x > 0$, then there exists $n \in \mathbb{N}$ with $nx > y$.

Proof. Suppose not, so $A = \{nx \mid n \in \mathbb{N}\}$ is bounded above by y . Let $\alpha = \sup A$. Since $x > 0$, the number $\alpha - x$ is not an upper bound, so $\alpha - x < mx$ for some $m \in \mathbb{N}$. Then $\alpha < (m+1)x \in A$, contradicting $\alpha = \sup A$. $\square$

Corollary 1.4

$\mathbb{Q}$ is dense in $\mathbb{R}$: if $x < y$ then there is $q \in \mathbb{Q}$ with $x < q < y$.

Remark 1.5. **Theorem 1.3** is exactly what fails in a non-Archimedean ordered field, e.g. the field of rational functions over $\mathbb{R}$.

1.2 Sequences

Definition 1.6. A sequence (a_n) **converges** to L if for every $\varepsilon > 0$ there is N such that $|a_n - L| < \varepsilon$ for all $n \geq N$.

Question 1.7. Why can a sequence not converge to two different limits?

Exercise 1.8. Show that every convergent sequence in $\mathbb{R}$ is bounded, and that the converse is false.

Proposition 1.9 (Monotone convergence)

Every bounded monotone sequence of reals converges.

Proof. Say (a_n) is increasing and bounded, and put $L = \sup_n a_n$. Given $\varepsilon > 0$, some $a_N > L - \varepsilon$; monotonicity gives

$$L - \varepsilon < a_N \leq a_n \leq L \quad \text{for all } n \geq N,$$

hence $|a_n - L| < \varepsilon$. $\square$

2 Problem set

Problem 2.1. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be continuous with $f(x) = 0$ for all $x \in \mathbb{Q}$. Prove $f \equiv 0$.

Solution. Fix $x \in \mathbb{R}$ and take rationals $q_n \rightarrow x$. Continuity gives $f(x) = \lim_n f(q_n) = 0$. $\square$